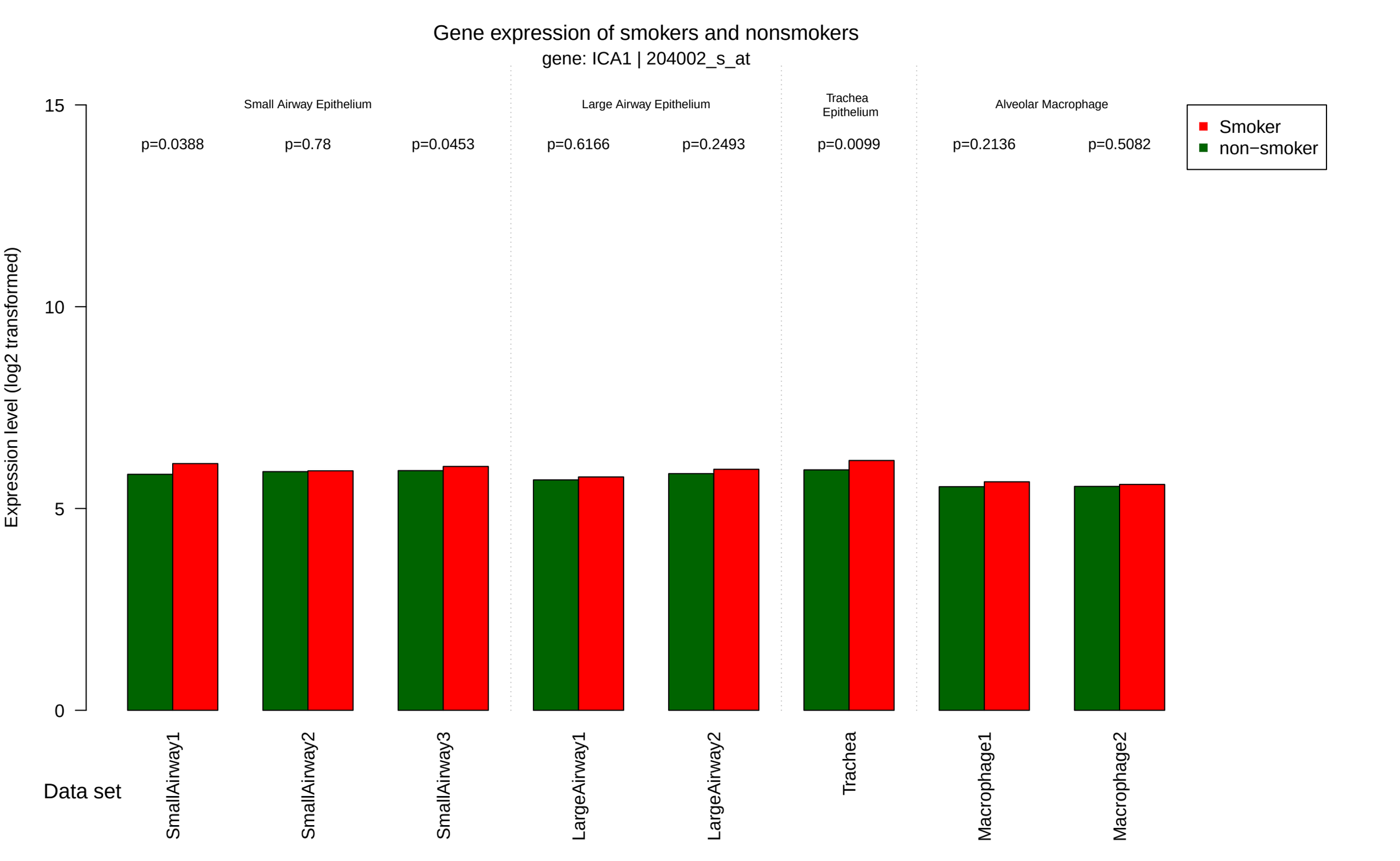
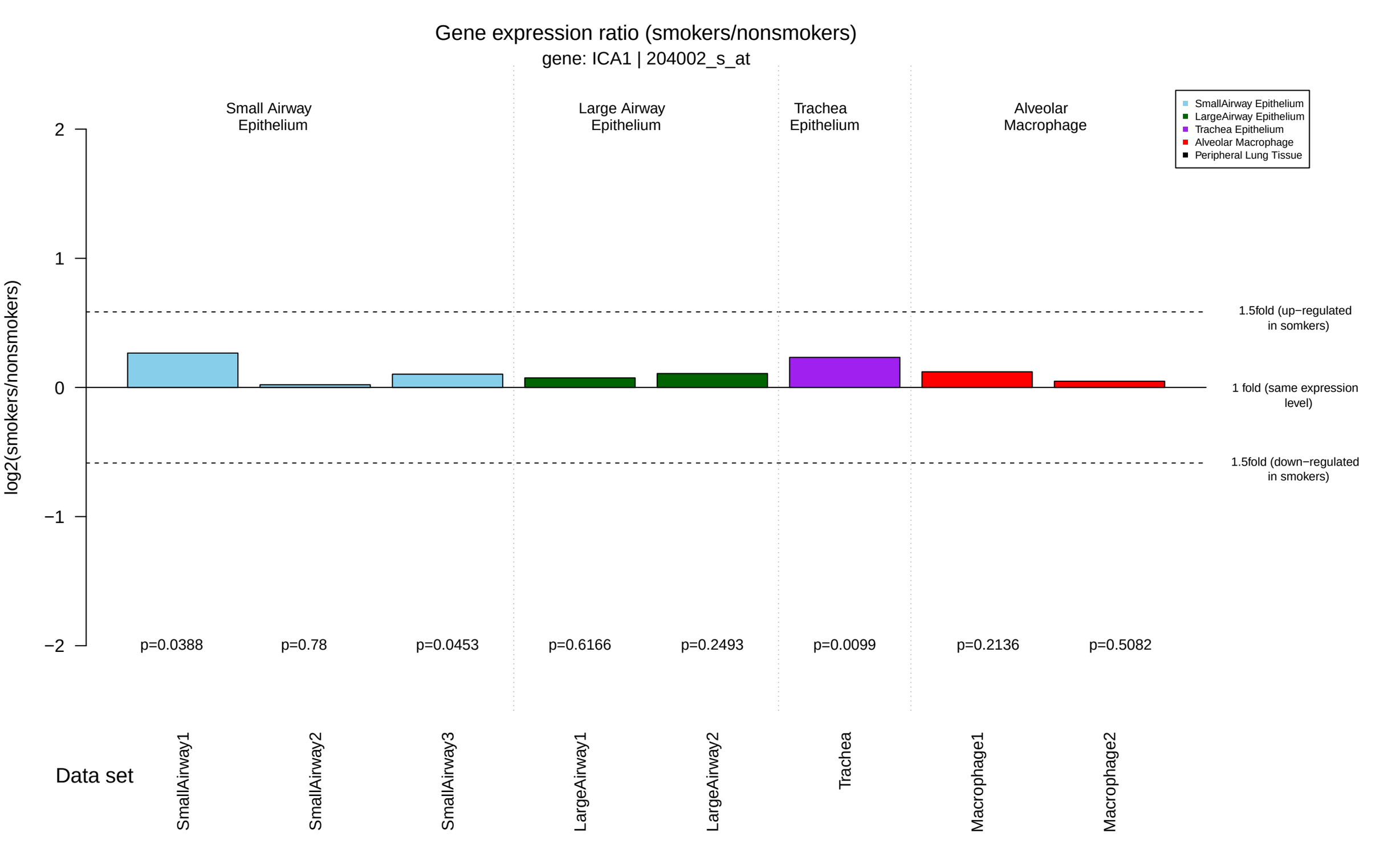
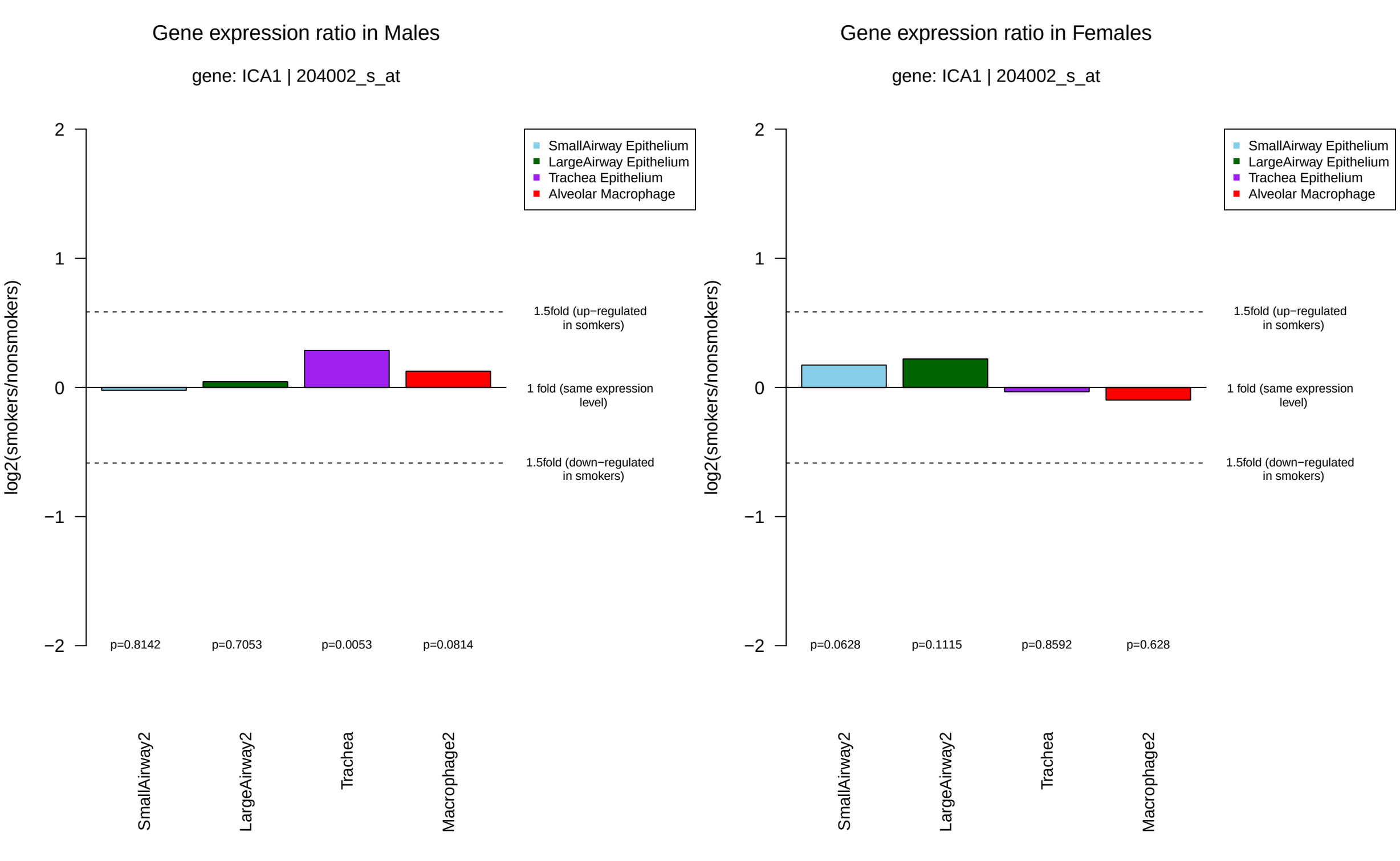
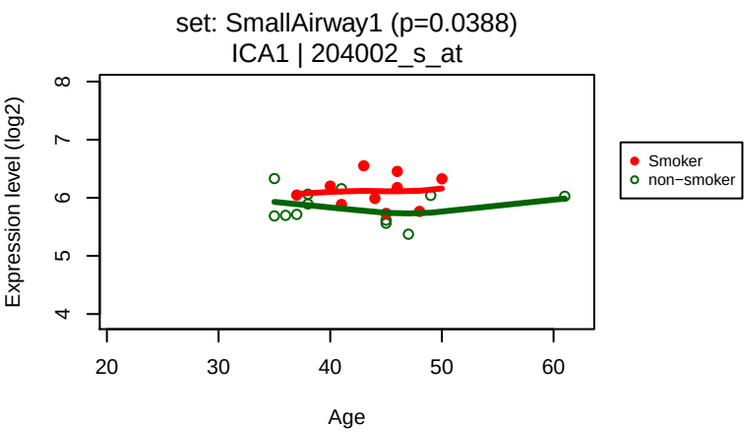
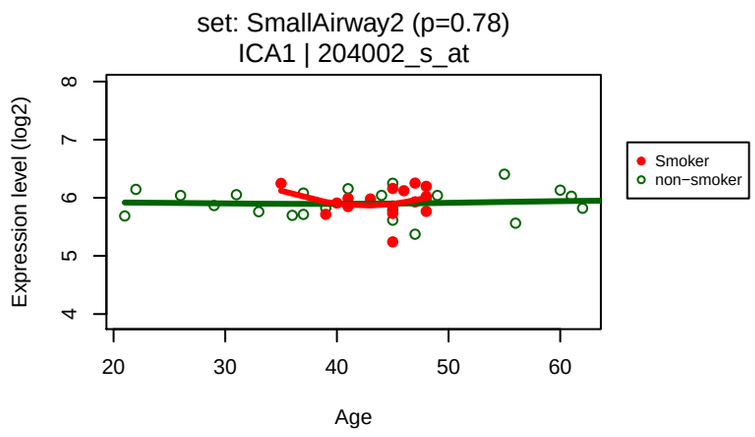
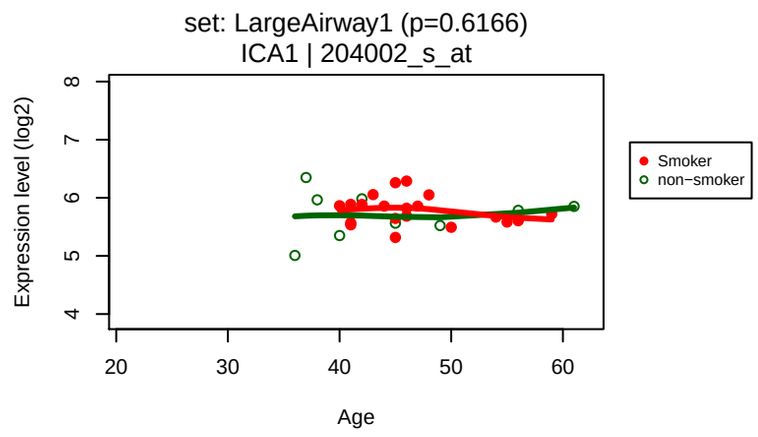

Gene expression of smokers and nonsmokers
gene: ICA1 | 204002_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0388
p=0.78
p=0.0453
p=0.6166
p=0.2493
p=0.0099
p=0.2136
p=0.5082
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

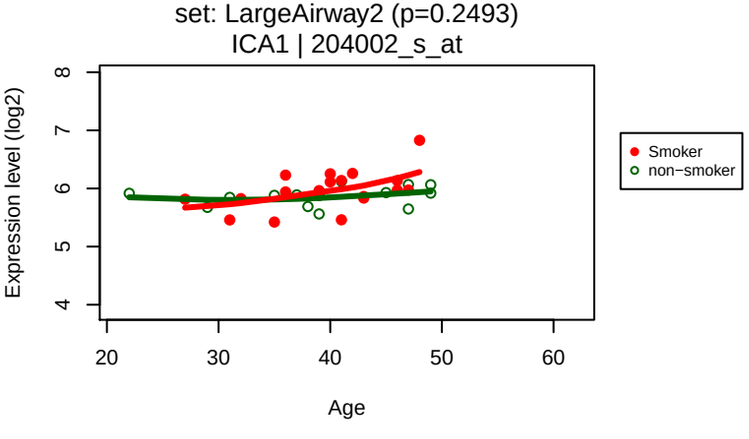

Gene expression ratio (smokers/nonsmokers)
gene: ICA1 | 204002_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0388
p=0.78
p=0.0453
p=0.6166
p=0.2493
p=0.0099
p=0.2136
p=0.5082
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

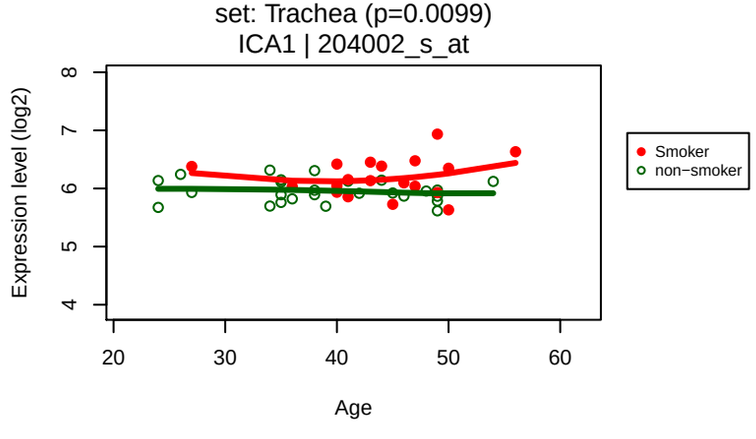

Gene expression ratio in Males
gene: ICA1 | 204002_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8142
p=0.7053
p=0.0053
p=0.0814
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ICA1 | 204002_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0628
p=0.1115
p=0.8592
p=0.628
SmallAirway2
LargeAirway2
Trachea
Macrophage2

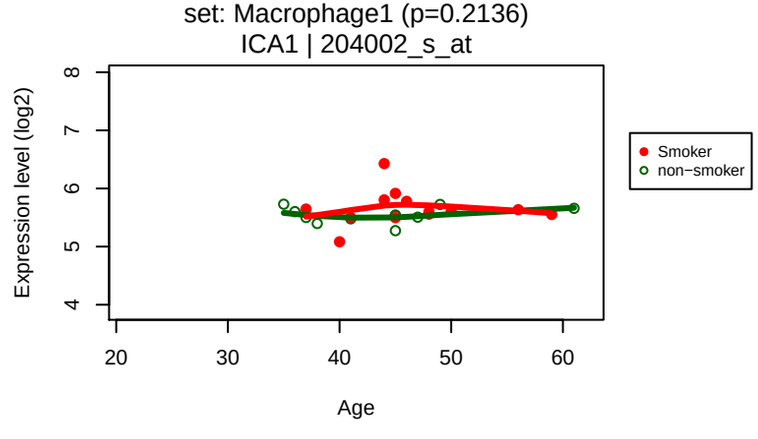

set: SmallAirway1 (p=0.0388)
ICA1 | 204002_s_at
Expression level (log2)
Age
Smoker
non−smoker

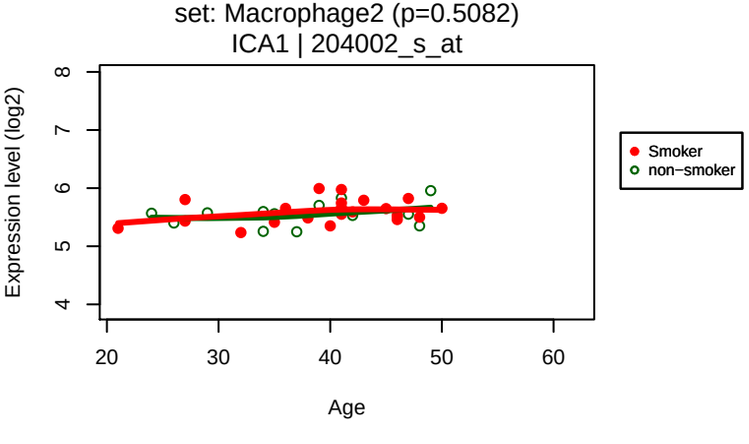

set: SmallAirway2 (p=0.78)
ICA1 | 204002_s_at
Expression level (log2)
Age
Smoker
non−smoker

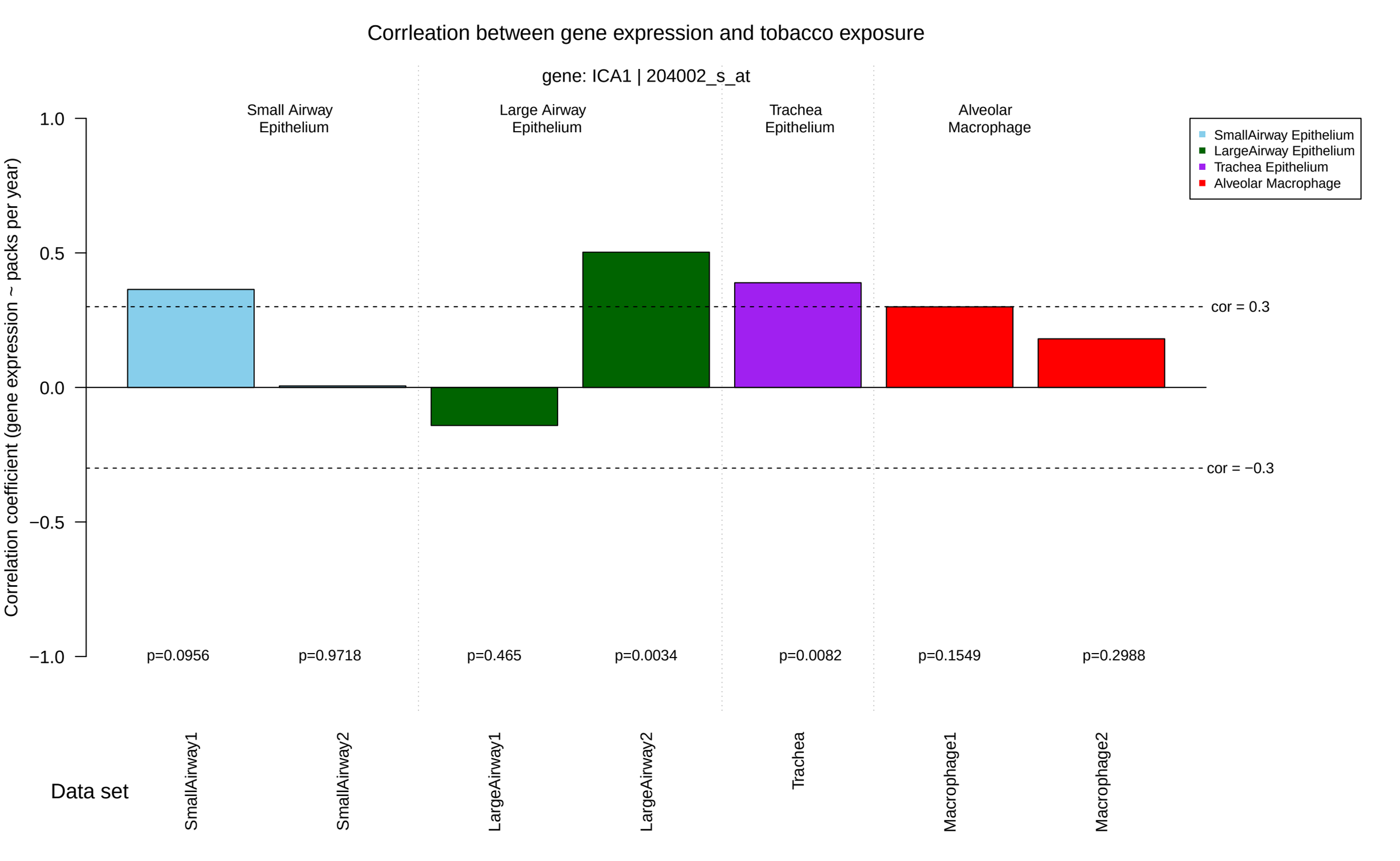

set: LargeAirway1 (p=0.6166)
ICA1 | 204002_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2493)
ICA1 | 204002_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0099)
ICA1 | 204002_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2136)
ICA1 | 204002_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5082)
ICA1 | 204002_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ICA1 | 204002_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0956
p=0.9718
p=0.465
p=0.0034
p=0.0082
p=0.1549
p=0.2988
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2